



POWER BI PORTFOLIO PROJECT

Texas Healthcare Workforce Access & Community Needs

County-level indicators for community health, shortage designations, workforce supply, facility access, and underserved areas

CDC PLACES 2024 • HRSA HPSA / MUA-P • AHRF

Project Objective

WHAT THE DASHBOARD ANSWERS

Where in Texas does community health need outpace the available healthcare workforce — and which counties should be prioritized first?

- Health need, shortage status, and workforce supply usually live in separate federal datasets.
- Planners and recruiters have no single county view to compare them side by side.
- This dashboard joins them on county FIPS so need and capacity are read together.
- The result supports directional prioritization for outreach, staffing, and grant targeting.



Executive Summary

THE HEADLINE FINDING

Border counties carry the sharpest health burden, while primary-care shortage designations extend across almost every Texas county.

Need clusters at the border

Fair-or-poor health reaches 48.8% versus a 24.1% Texas average; Kenedy, Zapata, and Dimmit lead.

Shortage is nearly statewide

942 primary-care HPSAs span 253 counties; 2,413 designations are reported across all three disciplines.

Capacity is uneven

Gaines reports 0.44 PCPs per 10K, and several counties report zero hospital beds.

Sources & Dashboard Architecture

THREE FEDERAL DATASETS, FOUR LINKED REPORT AREAS

01

Community Health

Four CDC PLACES pages on burden, behavior, status, and access

02

Shortage Designations

Primary care, dental, and mental health HPSA pages

03

Underserved Areas

MUA/P coverage and IMU-based underservice

04

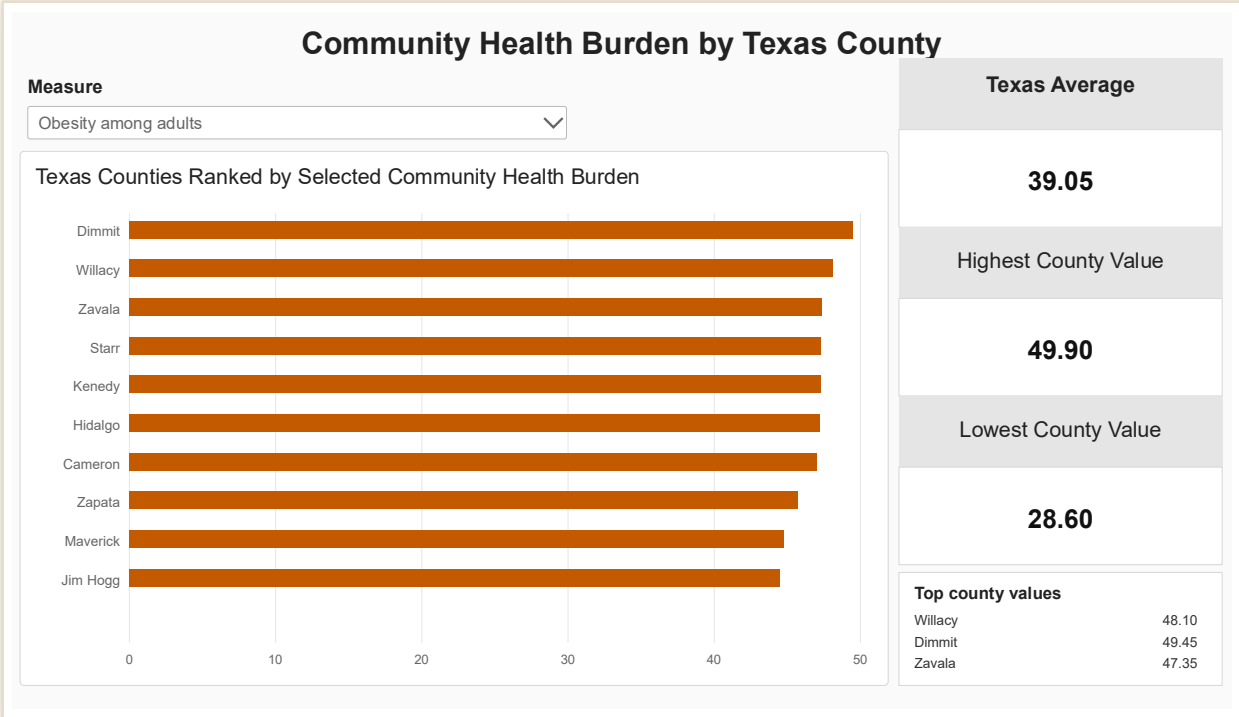
Workforce & Facilities

Two AHRF pages on supply, capacity, and benchmarks

CDC PLACES, HRSA HPSA / MUA-P, and AHRF are joined on county FIPS in Power Query — one row per Texas county across 11 analytical report pages, plus an overview and a methodology page.

Chronic disease burden concentrates along the Rio Grande border

SECTION 01 · CDC PLACES 2024 COUNTY DATA



Border belt

The highest-burden counties cluster along the Rio Grande border.

21-pt gap

Counties span 28.6% to 49.9% on adult obesity alone.

Measure-driven

Switching the slicer re-ranks all 254 counties instantly.

Need signal

High burden marks where demand on providers is greatest.

Section 02 — Shortage & Underserved

HRSA HPSA / MUA-P DATA

05

Primary Care

942 designations across 253 counties

06

Dental

724 designations across 241 counties

07

Mental Health

747 designations;
dashboard coverage is
virtually statewide

08–09

Summary & MUA/P

Discipline roll-up plus 165 underserved counties

Pages 05–09 cover designation counts, severity scores, shortage estimates, and MUA/P underservice.

Primary care shortage is effectively statewide — severity is the real filter

HRSA HPSA · SHORTAGE VIEW 1 OF 4

Primary Care Shortage Profile by Texas County

HPSA Designations

942

Counties with HPSA

253

Avg HPSA Score

11.82

Total Provider Shortage

37.10K

Provider shortage ranking

Bell	34.01
Webb	23.90
Bexar	21.16
El Paso	18.86
Orange	17.43
Starr	16.01
Hays	14.65
Hidalgo	13.16
Tarrant	12.78
Dallas	12.69

County detail

County	Avg HPSA	Provider shortage
Bell	17.00	34.01
Webb	15.33	23.90
Bexar	12.93	21.16
El Paso	15.68	18.86
Orange	9.00	17.43
Starr	15.63	16.01
Hays	17.00	14.65
Hidalgo	14.45	13.16
Tarrant	11.74	12.78
Dallas	13.44	12.69

Near-universal

253 of 254 counties carry a primary care shortage designation.

Scale

An estimated 37.1K providers short statewide.

Urban too

Bexar, Tarrant, and Dallas rank high despite metro size.

Score matters

Severity, not just designation status, drives prioritization.

Dental carries the steepest severity yet the narrowest footprint

HRSA HPSA · SHORTAGE VIEW 2 OF 4

Dental Shortage Profile by Texas County

HPSA Designations

724

Counties with HPSA

241

Avg HPSA Score

13.23

Total Provider Shortage

24K

Provider shortage ranking

Cameron		31.71
Bell		31.04
Nueces		30.72
Webb		26.48
Brazos		20.90
El Paso		18.08
Galveston		15.00
Hidalgo		14.90
Harris		14.59
Ector		14.01

County detail

County	Avg HPSA	Provider shortage
Cameron	13.00	31.71
Bell	14.00	31.04
Nueces	20.50	30.72
Webb	19.67	26.48
Brazos	21.33	20.90
El Paso	17.30	18.08
Galveston	12.50	15.00
Hidalgo	19.84	14.90
Harris	14.43	14.59
Ector	17.00	14.01

Steepest

Dental carries the steepest average HPSA score of the three.

Metro-weighted

Cameron, Bell, and Nueces top the shortage estimates.

Narrowest

241 designated — the narrowest footprint of the disciplines.

Often omitted

Dental access rarely appears in workforce planning.

Mental health shortage covers all 254 counties

HRSA HPSA · SHORTAGE VIEW 3 OF 4

Mental Health Shortage Profile by Texas County

HPSA Designations

747

Counties with HPSA

254

Avg HPSA Score

13.14

Total Provider Shortage

7.50K

Provider shortage ranking

Rockwall	58.40
Hidalgo	35.89
Ellis	31.33
Navarro	30.40
Cameron	21.22
Kaufman	20.19
Hunt	15.60
Travis	12.79
Willacy	12.02
Starr	11.22

County detail

County	Avg HPSA	Provider shortage
Rockwall	8.00	58.40
Hidalgo	13.00	35.89
Ellis	16.33	31.33
Navarro	16.33	30.40
Cameron	12.00	21.22
Kaufman	13.67	20.19
Hunt	14.60	15.60
Travis	11.00	12.79
Willacy	19.00	12.02
Starr	15.80	11.22

Statewide reach

747 designations span all 254 Texas counties in the report.

Supply floor

AHRF context reports 0.89 psychiatrists per 10K statewide.

Shortage extremes

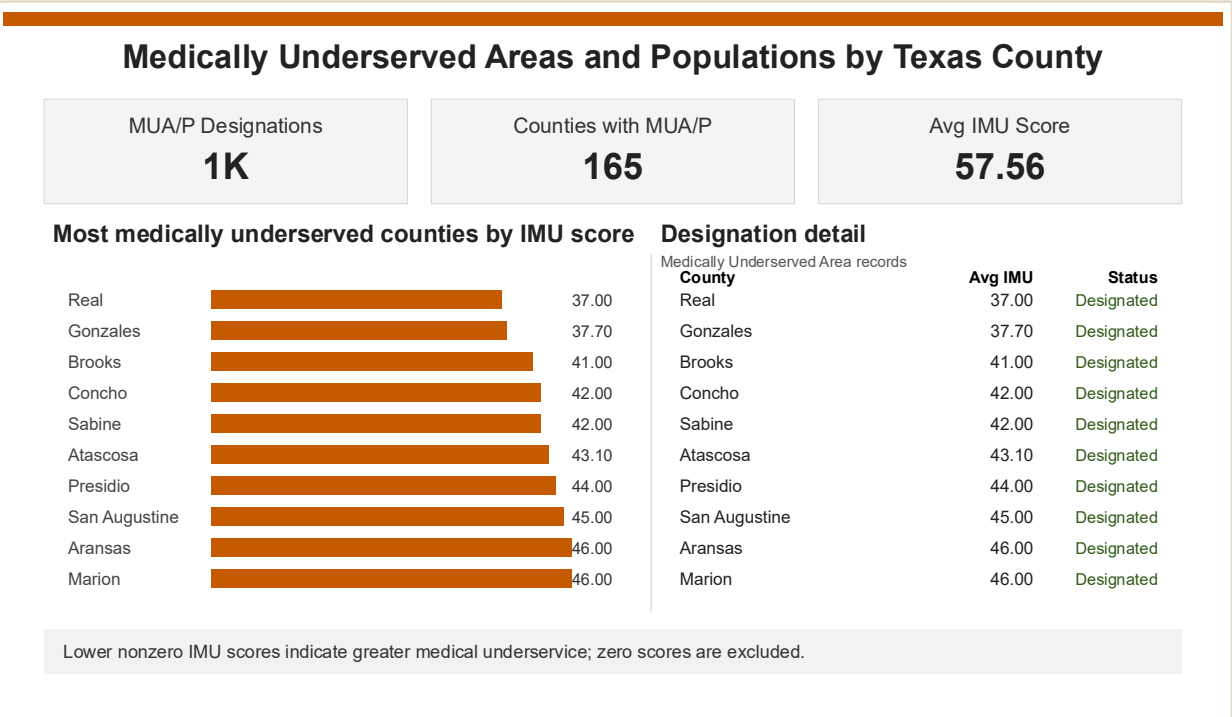
Rockwall (58.40), Hidalgo (35.89), and Ellis (31.33) lead the listed shortage estimates.

Complete coverage

The corrected PDF now shows the full 254-county Texas footprint.

Two-thirds of Texas counties are medically underserved

HRSA MUA-P · SHORTAGE VIEW 4 OF 4



Two-thirds

165 of 254 counties carry an MUA/P designation.

Lower is worse

IMU scores below the 57.6 average signal deeper underservice.

Real County

Lowest IMU score in the state at 37.0.

Rural skew

Designations concentrate in nonmetro and border counties.

Section 03 — Workforce & Facilities

AHRF COUNTY-LEVEL DATA

10

County Context

Provider supply, hospital capacity, and rural status

11

County Profile

Single-county view against Texas benchmarks

Per-10K Rates

All supply measures normalized by county population

Rural vs Metro

Access gaps split by rural-urban classification

Pages 10–11 compare workforce and facility capacity across 254 counties.

Provider supply and hospital capacity fall to zero in some counties

AHRF • WORKFORCE VIEW 1 OF 2

County Workforce & Facility Context by Texas County

Counties	Population	PCPs per 10K	Dentists per 10K	Psychiatrists per 10K	Beds per 10K
254	31M	5.93	6.32	0.89	25.08

Counties with lowest PCPs per 10K

Gaines		0.44
Robertson		0.57
Leon		0.60
San Jacinto		0.68
Shelby		0.83
Sabine		0.99
Parmer		1.03
Van Zandt		1.07
Waller		1.08
Archer		1.09

County workforce and facility detail

County	PCPs	Dentists	Beds	Rural category
Gaines	0.44	1.31	17.91	Nonmetro / Rural
Robertson	0.57	0.57	0.00	Metro
Leon	0.60	2.40	0.00	Nonmetro / Rural
San Jacinto	0.68	0.34	0.00	Metro
Shelby	0.83	2.89	0.00	Nonmetro / Rural
Sabine	0.99	2.98	24.86	Nonmetro / Rural
Parmer	1.03	1.03	25.86	Nonmetro / Rural
Van Zandt	1.07	2.14	0.00	Nonmetro / Rural
Waller	1.08	2.00	0.00	Metro
Archer	1.09	0.00	0.00	Metro

Provider and facility rates shown per 10,000 residents using AHRF county population estimates.
Lower values may indicate potential workforce or facility access gaps.

Supply floor

Some counties report under 1 primary care physician per 10K.

No beds

Several counties record zero hospital beds entirely.

Rural pattern

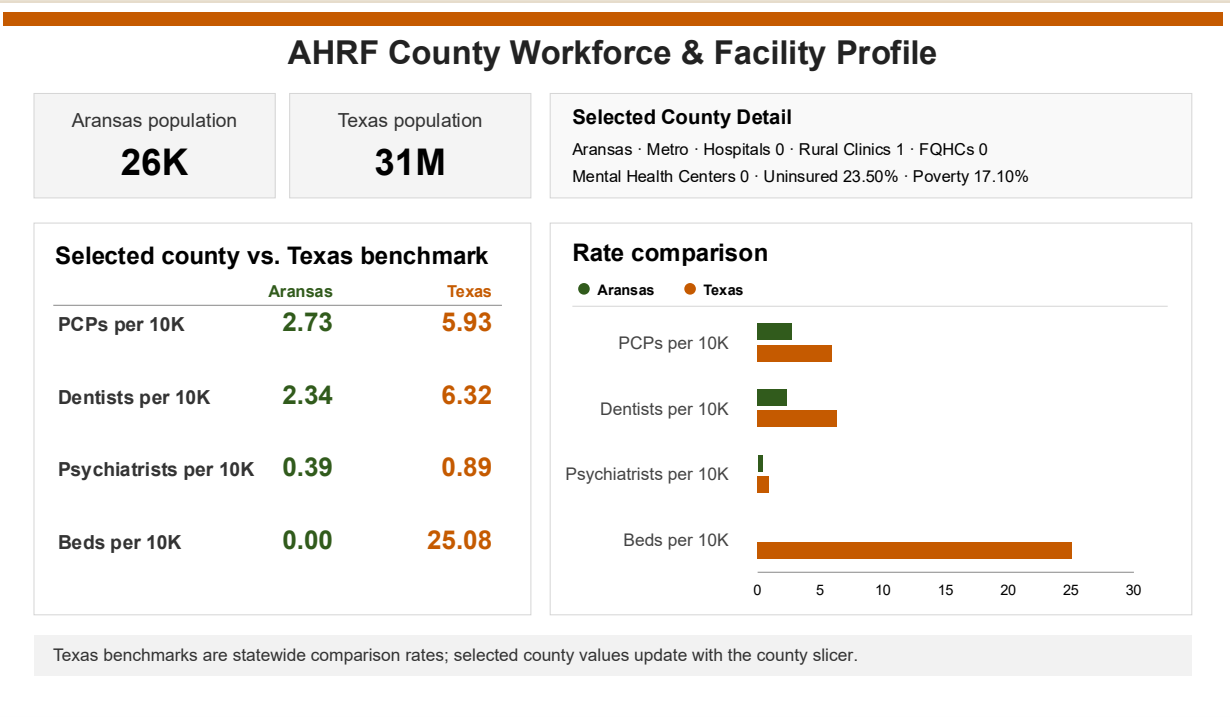
The lowest-supply counties are mostly nonmetro.

Capacity side

This is the supply half of the need-versus-capacity question.

County-level drill-down turns the analysis into a recruiting conversation

AHRF · WORKFORCE VIEW 2 OF 2



Drill-down

Any county can be measured against the state benchmark.

Aransas

Zero hospital beds against a 25.08 statewide rate.

Context added

Uninsured and poverty rates frame the access picture.

Recruiter view

Turns the analysis into a county-by-county conversation.

Key Findings

WHAT THE COUNTY VIEW REVEALS

01

Border counties recur

Kenedy, Dimmit, Zapata, Starr, Hidalgo, and Cameron lead on burden, behavior, and health status alike — inactivity spans 17.4% to 49.3%.

02

Use severity, not status alone

Primary-care HPSAs cover 253 counties; scores and shortage estimates separate the highest-priority markets.

03

Disciplines differ

Primary care has the largest shortage (37.10K); dental has the highest average HPSA score (13.23).

04

Layer facility capacity

County supply falls to 0.44 PCPs per 10K, and multiple counties report zero hospital beds.

Prevention runs opposite to need: colorectal screening spans 39.4% to 68.7% and peaks in suburban rather than border counties. MUA/P designations cover 165 counties, with Real County lowest at 37.0.

Priority Counties & Recommended Actions

TOP 10 COUNTIES, TIERED BY INTERVENTION

Recruit first

Gaines · Real · Aransas · Kenedy
Supply is the binding constraint:
Gaines 0.44 PCPs per 10K, Real
lowest IMU at 37.0, Aransas zero
hospital beds.

Grant-target

Zapata · Dimmit · Willacy
Highest burden plus MUA/P
designation. Fair-or-poor health hits
48.8% against a 24.1% state average.

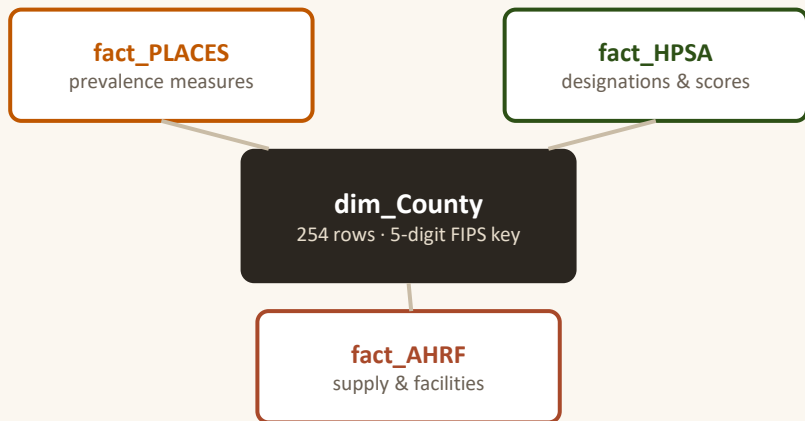
Telehealth scale

Hidalgo · Cameron · Starr
Large populations, severe mental-
health shortage (Hidalgo 35.89),
existing infrastructure to build on.

Tiers weight four inputs equally — health-burden rank, HPSA severity score, IMU score, and PCPs per 10K. Counties in the worst quartile on supply go to Recruit; worst on burden with MUA/P status go to Grant-target; high-population counties with severe shortage but existing infrastructure go to Telehealth.

How the Model Was Built

DATA MODEL, TRANSFORMATION, AND MEASURE DESIGN



Power Query cleanup

Filter to Texas, pad FIPS to five digits, unpivot PLACES measures, normalize county names, roll HPSA rows to one per county.

Key DAX patterns

DIVIDE with a zero guard for per-10K rates, RANKX leaderboards, variance to an ALL-county benchmark, and a SWITCH measure selector.

One county grain across all three sources keeps every page comparable — the same slicer drives health, shortage, and workforce views without bridging tables.

How the Numbers Were Validated

JOIN INTEGRITY, COVERAGE, AND RECONCILIATION

Join integrity

Row counts were checked at each merge step so that all three sources resolve to a single set of 254 county rows without duplicates or unmatched keys.

Coverage checks

County coverage was reviewed against the source extracts; a missing-county gap in the mental health view was traced and corrected.

Known discrepancies

Where sources disagree — such as differing statewide psychiatrist rates — the conflict is surfaced on the methodology page rather than averaged away.

FIPS padding, type casting, and county-count checks are part of the query steps, so a change in a source extract shows up as an error rather than a quiet shift in the numbers.

Limitations & Methodology

HOW TO READ THE RESULTS

Directional, not clinical

CDC PLACES measures are modeled county estimates, not exact clinical counts. Rates in very small counties such as Kenedy swing widely on a handful of people.

Different vintages

Source years, definitions, and HRSA designation methods differ. HPSAs are also designated for sub-county areas, populations, and facilities, then rolled up here to a county grain.

Access is broader

Rates do not capture transportation, insurance, appointment availability, or within-county distribution — and residents routinely cross county lines for care.

Portfolio Summary

WHAT THIS PROJECT DEMONSTRATES

An end-to-end analytics build: three public datasets joined into one county model and delivered as a 13-page export covering all 254 Texas counties. Sources: CDC PLACES · HRSA HPSA / MUA-P · AHRF.

Technical depth

Query transformation, star-schema modeling, and DAX measure design end to end.

Domain fluency

Comfort with HPSA, MUA-P, and prevalence measures as analytic inputs.

Decision-ready output

Prioritization insights paired with transparent methods, limitations, and QA checks.